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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Power Electronics Laboratory

Final Project Report

Section: A1

Group: 02

Design and Hardware Implementation of a
12V to 5V and 12V to 18V
Buck-Boost DC-DC Converter

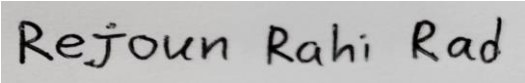
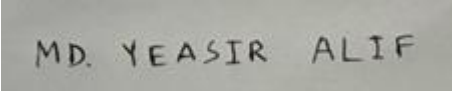
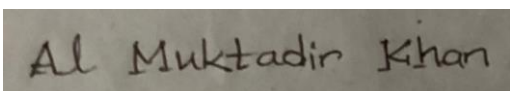
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1. Abstract

This project presents the design, simulation, and hardware implementation of a DC–DC buck–boost converter capable of both stepping down and stepping up an input voltage. The system is designed to convert a fixed 12V DC input into two regulated output levels: 5V (buck mode) and 18V (boost mode). Pulse Width Modulation (PWM) control is employed to regulate the output voltage by adjusting the duty cycle of the switching signal.

The design process includes theoretical calculations of duty cycle, inductor, and capacitor values, followed by simulation using Proteus and practical hardware implementation. Performance parameters such as output voltage, power losses, and efficiency are analyzed and compared between ideal and experimental results. The hardware prototype demonstrates successful operation with reasonable efficiency, validating the effectiveness of the proposed design.

2. Introduction

DC–DC converters are essential components in modern electronic systems where voltage level conversion is required for proper operation of devices. Among various topologies, the buck–boost converter is particularly important due to its ability to both increase and decrease the input voltage magnitude using a single circuit configuration. This makes it highly suitable for applications such as battery-powered systems, renewable energy systems, and embedded electronics where input voltage may vary.

In this project, a buck–boost converter is designed to operate with a 12V input supply and provide two different regulated outputs: 5V for low-voltage applications and 18V for higher voltage requirements. The system uses a MOSFET-based switching mechanism controlled by PWM signals to achieve the desired voltage regulation.

This problem qualifies as a complex engineering problem because it involves nonlinear switching behavior, dynamic system response, component non-idealities, and trade-offs between efficiency, ripple, and stability. Additionally, practical implementation introduces challenges such as switching losses, inductor design constraints, and thermal considerations.

Alternative solutions to this problem include using separate buck and boost converters, linear regulators, or more advanced topologies such as SEPIC or Cuk converters. However, the buck–boost converter provides a compact and cost-effective solution with fewer components, making it an efficient choice for this application.

3. Design

3.1 Problem Formulation (PO(b))

3.1.1. Identification of Scope

The scope of this project includes the design and implementation of a DC–DC buck–boost converter that can operate in both buck and boost modes. The system is required to:

- Accept a fixed DC input of 12V
- Produce regulated outputs of 5V (buck mode) and 18V (boost mode)
- Operate using PWM-based switching at a frequency of 40 kHz
- Maintain acceptable levels of voltage ripple (approximately 1%)
- Deliver power to a resistive load of 10 Ω

The project covers theoretical design, simulation, hardware implementation, and performance evaluation.

3.1.2 Literature Review

Buck–boost converters have attracted extensive interest in both research and industrial domains

because of their ability to provide an output voltage that can be either higher or lower than the input. The body of literature on buck–boost converters spans core operating concepts, advanced control strategies, performance enhancement, compact implementation, and practical deployment across modern electronic systems.

A. Fundamental Principles

Initial studies on buck–boost converters established the theoretical framework describing their dual-mode voltage conversion capability. In *Power Electronics: Devices, Circuits, and Applications (4th Edition)* by Muhammad H. Rashid [1], the operating mechanisms, switching states, and steady-state as well as transient behaviors of buck–boost converters are analyzed in detail. These works emphasize inductor energy storage, polarity inversion (in traditional topologies), and the relationship between duty cycle and output voltage.

B. Control Techniques

To ensure stable and precise output regulation, a variety of control methods have been explored. Pulse Width Modulation (PWM) remains a dominant technique, allowing dynamic adjustment of the duty cycle to maintain desired output levels. Rashid [1] discusses classical PWM-based control along with feedback loop design. More recent literature also investigates current-mode control, voltage-mode control, and digital control strategies to improve dynamic response and robustness under varying load and input conditions.

C. Efficiency Optimization

Improving efficiency is a central concern in buck–boost converter design due to switching and conduction losses. Research has focused on techniques such as synchronous rectification, soft-switching methods (e.g., zero-voltage and zero-current switching), and optimized component selection. These approaches, discussed in standard texts and modern studies, aim to reduce power dissipation and enhance performance, particularly in low-power and battery-operated systems.

D. Integration and Miniaturization

With advancements in semiconductor fabrication and power IC design, buck–boost converters have increasingly been integrated into compact, highly efficient modules. In *Power Electronics* by Daniel W. Hart [2], integrated solutions are highlighted, showing how modern designs combine control circuitry, switching devices, and passive components into single-chip or system-in-package configurations. This trend is especially relevant for portable electronics, wearable devices, and embedded systems where space and power efficiency are critical.

E. Applications and Case Studies

Buck–boost converters are widely used in applications requiring flexible voltage regulation, such as battery-powered devices, renewable energy systems, and automotive electronics. Both Rashid [1] and Hart [2] provide examples of their use in solar energy systems (to handle variable input voltages), electric vehicles (for battery management), and communication systems. These case studies demonstrate the versatility of buck–boost converters in maintaining stable operation across fluctuating conditions.

Overall, the literature reflects continuous advancements aimed at improving the adaptability, efficiency, and integration of buck–boost converters, ensuring their relevance in next-generation electronic and energy systems.

3.1.3 Formulation of Problem

Given specifications:

- Input voltage: $V_s = 12\text{V}$
- Output voltages:
- Buck mode: $V_o = 5\text{V}$

- Boost mode: $V_o = 18V$
- Load resistance: $R = 10\Omega$
- Switching frequency: $f_s = 40kHz$
- Allowable ripple: approximately 1%

The converter must operate in two modes:

- 1 Buck Mode (Step-down): Convert 12V to 5V
- 2 Boost Mode (Step-up): Convert 12V to 18V

The design problem involves determining:

- Appropriate duty cycle D for both modes
- Minimum inductance L to ensure continuous conduction mode (CCM)
- Minimum capacitance C to maintain low output voltage ripple
- Expected inductor current ripple and overall system performance

The system must ensure stable output voltage, acceptable ripple, and efficient operation under both modes.

3.1.4 Analysis

A buck–boost converter is a type of DC–DC power converter that can produce an output voltage either lower or higher than the input voltage by controlling the switching behavior of the circuit. It operates using an inductor, a switching device (MOSFET), a diode, and an output capacitor. The fundamental principle behind its operation is the controlled storage and release of energy in the inductor.

The working of the buck–boost converter can be understood in two main switching states within each cycle:

1. Switch ON State:

When the MOSFET switch is turned ON, the input voltage is directly applied across the inductor. During this period, the diode becomes reverse biased, and no energy is transferred to the output. Instead, energy is stored in the magnetic field of the inductor as current through it increases gradually. The load is temporarily powered by the output capacitor, which supplies energy during this interval.

2. Switch OFF State:

When the MOSFET is turned OFF, the inductor opposes the sudden decrease in current by reversing its polarity. This causes the diode to become forward biased, allowing the stored energy in the inductor to be transferred to the load and the output capacitor. The inductor current decreases as it delivers energy to the output.

By continuously switching between these two states at a high frequency, the converter regulates the output voltage. The key control parameter is the duty cycle, which represents the fraction of time the switch remains ON during each switching period.

- If the duty cycle is low, less energy is stored in the inductor, resulting in a lower output voltage (buck operation).
- If the duty cycle is high, more energy is stored and transferred, resulting in a higher output voltage (boost operation).

In this project, the converter is required to operate in both modes using the same circuit. Therefore, careful control of the duty cycle is essential to achieve the desired output voltages.

From a design perspective, the main objective is to develop a system that can reliably perform this

energy conversion while maintaining stability and efficiency. Specifically, the following aspects must be designed and optimized:

- **Duty Cycle Control:**

The PWM signal must be generated such that it produces approximately 30% duty cycle for buck operation (5V output) and around 60% duty cycle for boost operation (18V output).

- **Inductor Design:**

The inductor must be selected to ensure continuous current flow and to limit current ripple. An improperly chosen inductor can lead to discontinuous conduction or excessive losses.

- **Capacitor Selection:**

The output capacitor must smooth the voltage and maintain ripple within acceptable limits (around 1%). It also plays a crucial role in supplying the load during switching transitions.

- **Switching Device Performance:**

The MOSFET must handle high-frequency switching efficiently with minimal losses, while the diode must provide fast recovery and low forward voltage drop.

- **System Efficiency:**

Losses due to switching, resistance, and magnetic components must be minimized to ensure acceptable efficiency in both operating modes.

In summary, the analysis of the buck–boost converter involves understanding the dynamic energy transfer between input, inductor, and output, and designing the system components and control strategy such that the required voltage conversion (12V to 5V and 12V to 18V) is achieved with acceptable ripple, stability, and efficiency.

3.2 Design Method (PO(a))

Mathematical calculations:

Duty Cycle Calculation

$$D = \frac{V_o}{V_o + V_s}$$

For buck operation,

$$D = \frac{5}{5 + 12} = 0.294 \approx 0.3$$

For boost operation,

$$D = \frac{18}{18 + 12} = 0.6$$

Inductor Design

To ensure continuous conduction mode (CCM), the minimum inductance is given by:

$$L_{min} = \frac{R(1 - D)^2}{2f_s}$$

For buck,

$$L_{min} = \frac{10(1 - 0.3)^2}{2 \times 40000}$$

$$= 61.25 \mu H$$

For boost,

$$L_{min} = \frac{10(1 - 0.6)^2}{2 \times 40000} = 20 \mu H$$

Capacitor Design

The output capacitor is selected based on voltage ripple constraint:

$$C_{min} = \frac{D}{R \times r \times f_s}$$

For buck,

$$C_{min} = \frac{0.3}{10 \times 0.01 \times 40000} = 75 \mu F$$

For boost,

$$C_{min} = \frac{0.6}{10 \times 0.01 \times 40000} = 150 \mu F$$

Inductor Current Ripple

The inductor ripple current is given by:

$$\Delta I_L = \frac{V_s D}{L f_s}$$

For buck,

$$\Delta I_L = \frac{12 \times 0.3}{61.25 \times 40000 \times 10^{-6}} = 1.47$$

For boost,

$$\Delta I_L = \frac{12 \times 0.6}{61.25 \times 10^{-6} \times 40000} = 2.94$$

Output Current

The output current is:

$$I_o = \frac{V_o}{R}$$

Buck mode:

$$I_o = \frac{5}{10} = 0.5$$

Boost mode:

$$I_o = \frac{18}{10} = 1.8$$

Power Calculation

$$P_{in} = P_{out} = V_o I_o$$

Buck mode:

$$P_{in} = 5 \times 0.5 = 2.5$$

Boost mode:

$$P_{in} = 18 \times 1.8 = 32.4$$

The following components are utilized in the circuit:

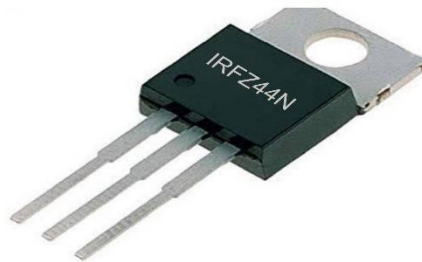
- Switching Element (MOSFET – IRFZ44N, rating-55V, 48A)

The switching element is a critical component in a buck–boost converter as it controls the energy transfer between the input and output. In this project, the IRFZ44N MOSFET is used due to its suitability for medium-power, high-frequency switching applications.

The IRFZ44N is an N-channel enhancement-type MOSFET capable of handling high current (up to 48A) and moderate voltage levels (up to 55V). Its low drain-to-source ON resistance ($R_{ds(on)}$) ensures reduced conduction losses during operation, which is essential for improving overall efficiency.

In the buck–boost converter, the MOSFET operates as a high-speed switch controlled by a PWM signal. During the ON state, it allows current to flow through the inductor, enabling energy storage. During the OFF state, it blocks current flow, forcing the inductor to transfer stored energy to the output through the diode.

The performance of the MOSFET directly affects switching losses, efficiency, and thermal behavior. Therefore, selecting a MOSFET with fast switching characteristics, low gate charge, and low ON resistance is crucial. The IRFZ44N meets these requirements reasonably well for the given operating frequency of 40 kHz, making it a practical and cost-effective choice for this design.



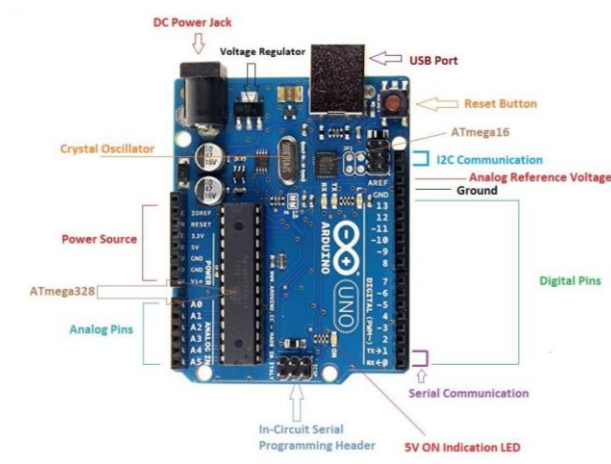
- Microcontroller (Arduino)

The microcontroller plays a vital role in generating the control signal required for switching the MOSFET. In this project, an Arduino microcontroller is used to generate Pulse Width Modulation (PWM) signals with adjustable duty cycles.

The Arduino provides a flexible and programmable platform for controlling the converter operation. By varying the duty cycle of the PWM signal, the output voltage of the buck–boost converter can be regulated. For this design, two duty cycle ranges are used: approximately 30% for buck operation and 60–65% for boost operation.

One of the major advantages of using Arduino is its ease of programming and real-time control capability. It allows rapid prototyping and testing without requiring complex hardware design. Additionally, the microcontroller can be further extended to implement closed-loop control systems by incorporating feedback from the output voltage.

The PWM signal generated by the Arduino, however, is not sufficient to directly drive the MOSFET due to limited current sourcing capability. Therefore, a MOSFET driver circuit is required to interface between the Arduino and the switching device.



- MOSFET Driver IC (MIC4422)

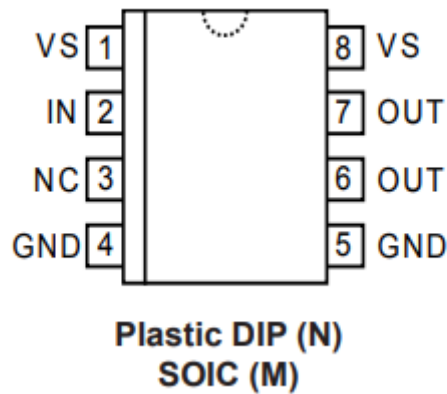
To ensure proper switching of the MOSFET, a dedicated gate driver IC, MIC4422, is used in this design. The MIC4422 is a high-speed, high-current MOSFET driver capable of delivering large peak currents (up to 9A), which is necessary for fast charging and discharging of the MOSFET gate capacitance.

The Arduino output alone cannot provide sufficient current to switch the MOSFET efficiently at high frequency. If directly connected, it would result in slow switching transitions, increased switching losses, and excessive heating. The MIC4422 solves this problem by acting as an interface between the low-power control signal and the high-power switching device.

When the PWM signal from the Arduino is applied to the input of the MIC4422, it amplifies the signal and provides a strong gate drive voltage to the MOSFET. This ensures rapid switching between ON and OFF states, minimizing transition time and reducing power loss.

Additionally, the driver IC improves system reliability by providing better noise immunity and consistent switching performance. This is particularly important in power electronic circuits where high-frequency switching can introduce significant electromagnetic interference.





Pin Description

Pin Number TO-220-5	Pin Number DIP, SOIC	Pin Name	Pin Function
1	2	IN	Control Input
2, 4	4, 5	GND	Ground: Duplicate pins must be externally connected together.
3, TAB	1, 8	V_s	Supply Input: Duplicate pins must be externally connected together.
5	6, 7	OUT	Output: Duplicate pins must be externally connected together.
	3	NC	Not connected.

- Input DC Voltage Source

The input supply for the buck–boost converter is provided using a regulated laboratory DC power supply. This source delivers a constant 12V DC input to the circuit, which serves as the baseline voltage for both buck and boost operations.

Using a laboratory power supply ensures stable and controllable input conditions during testing and experimentation. It allows precise adjustment and monitoring of voltage and current, which is essential for analyzing converter performance, efficiency, and response under different operating conditions.

- Load Resistance (10 Ω , 50W, Aluminium Housing Resistor)

The load in this project is implemented using a 10 Ω , 50W aluminium housing resistor, which acts as a resistive load for the converter output.

The high power rating (50W) ensures that the resistor can safely dissipate heat generated during operation, especially in boost mode where output power is significantly higher.

The aluminium housing provides effective heat dissipation, preventing overheating and maintaining safe operation. This is particularly important because resistive loads can experience substantial temperature rise during prolonged operation.



Using a fixed resistive load also simplifies analysis, as it allows straightforward calculation of output current and power using Ohm's law, making it easier to evaluate converter performance.

- Schottky Diode (SR560, rating-60V,5A)

A Schottky diode (SR560) is used in the converter due to its fast switching characteristics and low forward voltage drop.

In the buck–boost converter, the diode plays a crucial role during the OFF state of the MOSFET. When the switch turns OFF, the inductor releases its stored energy, and the diode provides a path for current to flow to the output. Therefore, the diode must respond quickly to switching transitions.

Schottky diodes are preferred over conventional PN junction diodes because they have:

- Lower forward voltage drop, which reduces conduction losses
- Faster switching speed, minimizing reverse recovery losses
- Better efficiency at high frequencies

The SR560 diode, rated at 60V and 5A, is suitable for the voltage and current levels in this project. Its use contributes to improved efficiency and reliable operation of the converter.

- Inductor (Iron Powder Toroidal Core – T141-52)

The inductor is a key energy storage element in the buck–boost converter. In this project, an iron powder toroidal core (T141-52 model) is used to construct the inductor.

Considering the high frequency we are using it is the most suitable one we have found available. At switching frequency, the core is repeatedly magnetized and demagnetized. Ordinary iron core would have much higher loss. Ferrite/iron powder wastes less energy as heat.

The toroidal shape provides several advantages, including reduced electromagnetic interference (EMI), compact size, and efficient magnetic flux containment. Iron powder cores are particularly suitable for power electronics applications because they offer distributed air gaps, which help prevent core saturation and allow stable operation under varying current conditions.

The number of turns is calculated based on the core's inductance factor (A_L) and desired inductance value. Proper winding and core selection are essential to minimize losses due to resistance and magnetic effects.

Additionally, the inductor must handle the peak current without saturating, especially during boost operation where current levels are higher.

Our target is to make an inductor of 61.25 μ F with current rating of 5A.

Technical Specifications:

Iron Powder Core Toroid

Model: T141-52

Material: Iron

Color: Blue-Green

External Diameter: approx. 36mm

Inside Diameter: approx. 22.4mm

Thickness: approx. 10.5mm

Inductance coefficient : A_L 95nH/N²

Permeability: 75 μ o approximately

Turns calculation:

$$L = A_L \cdot N^2$$

$$61250 = 95 \cdot N^2$$

So, N is approximately 26 turns

Safe operating current density,

$$J = 4 \text{ A/mm}^2$$

Required copper area,

$$A = I/J = 5/4 = 1.25 \text{ mm}^2$$

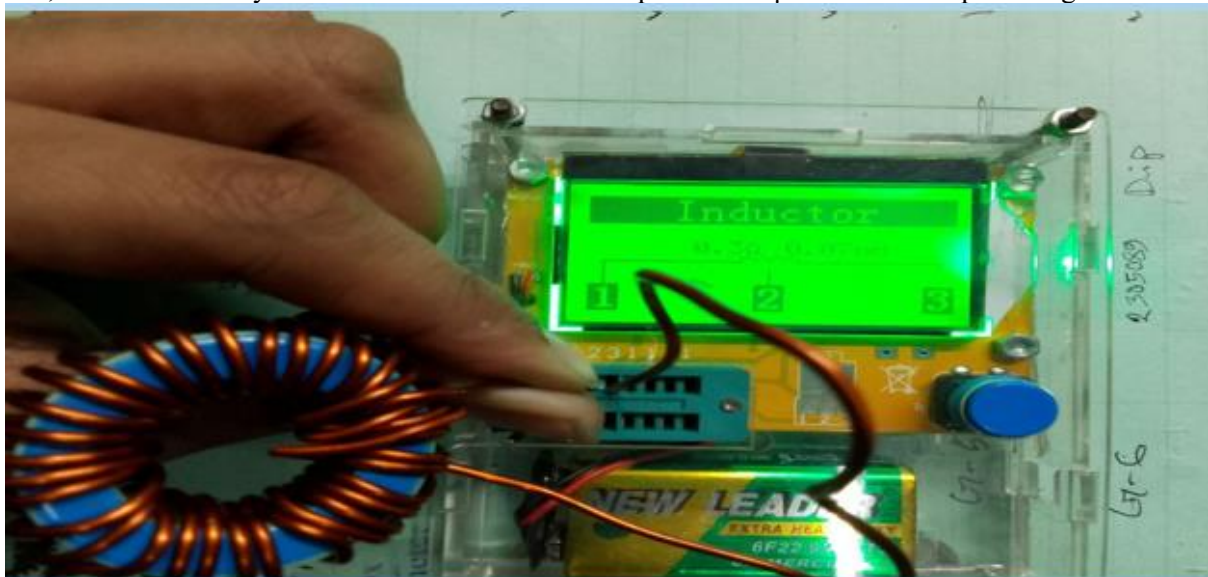
We select 16 SWG copper wire.

It has diameter of 1.63mm and area 2.08 mm².

Max safe current, $I_{\max} = 4 \cdot 2.08 = 8.3 \text{ A}$

Which satisfies our requirement.

But, when we actually tried to build the inductor we opted for 70 μ H and ended up needing 34 turns.



As we can see from the picture, the inductance is 0.07 mH and the series resistance because of the wire has come out to be 0.3 ohm.

- Capacitor

A 470 μ F capacitor, is connected across the output to filter out voltage ripple and stabilize the output voltage. It provides smooth and regulated DC voltage to the load.

The voltage rating of the capacitor is **50V**.

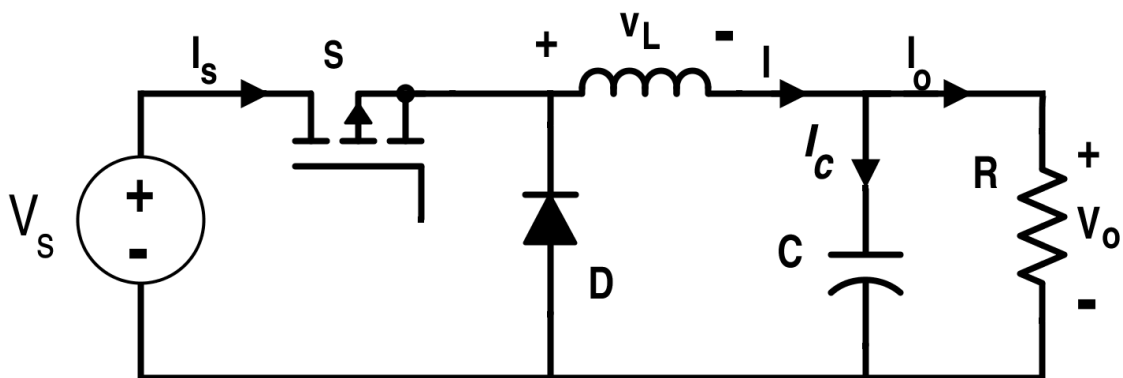
- 10 AWG wire:

As our buck-boost operates with high current in the boost mode, we can't use normal jumper wires to connect the components.



So, we had to use wire with high current rating. We selected 10 AWG with current rating 30A.

3.3 Circuit Diagram

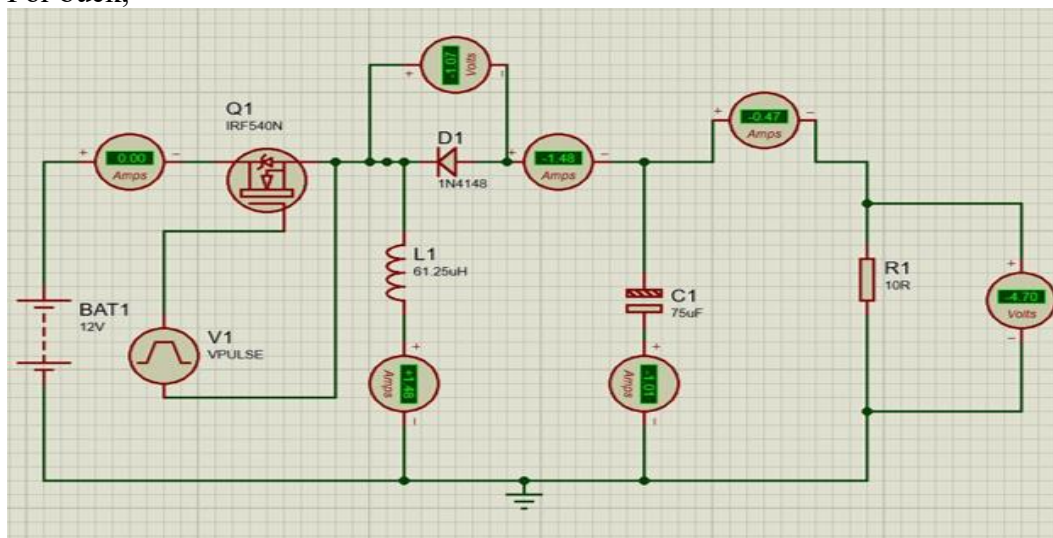


3.4 Simulation Model

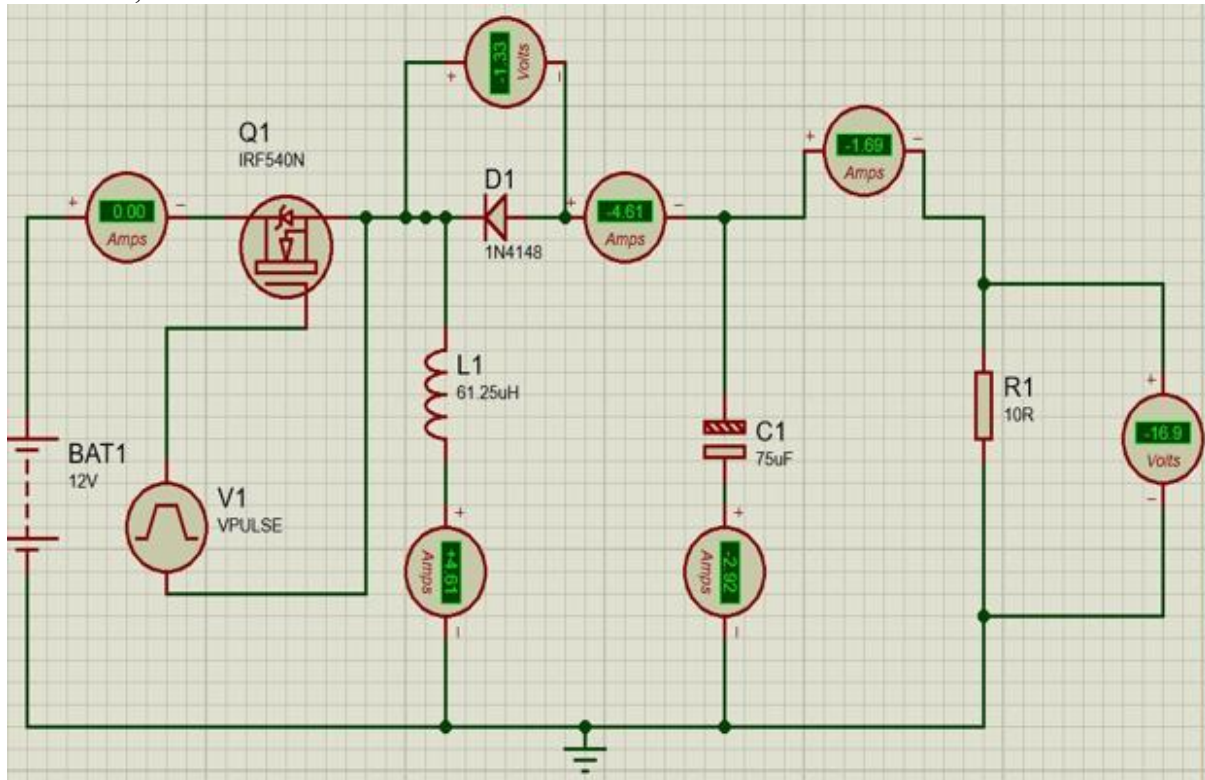
We used Proteus simulation software to check the output of our circuit for both buck and boost operation.

Initially, we used the critical values for inductor and capacitor, found from the theoretical calculation.

For buck,



For Boost,

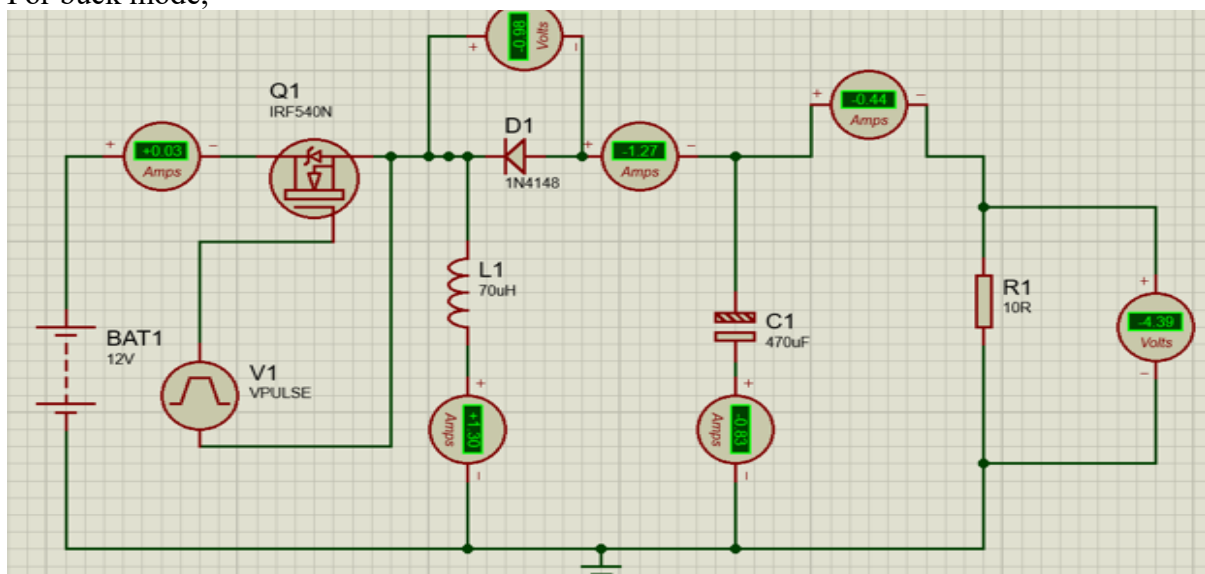


As we can see from the output,
We have got the correct operation in each mode although the output voltage is a bit lower than our target.

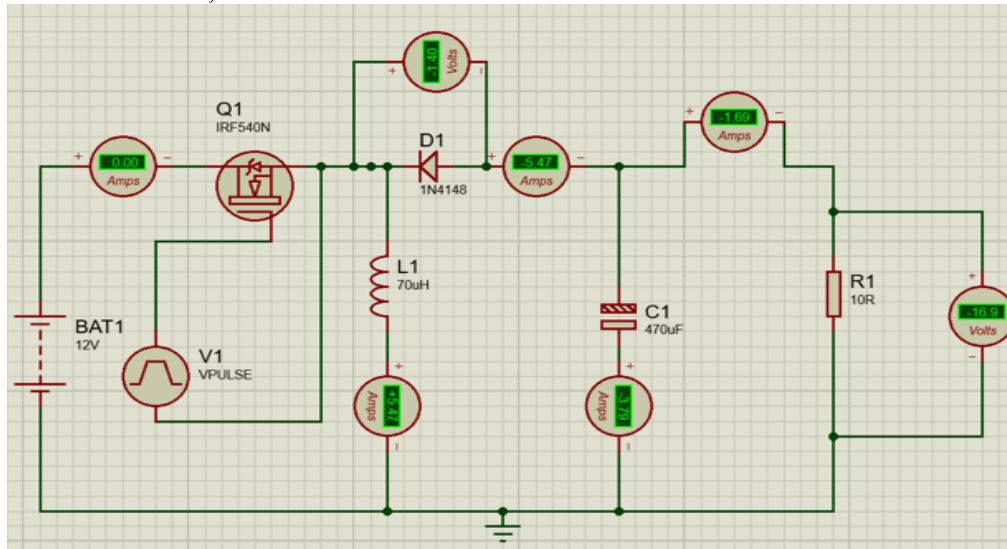
However, in physical experiment we used different values of inductor and capacitor because of availability and ease of designing.

So, next we did the simulation with inductance value $70\ \mu\text{H}$ and capacitance value $470\ \mu\text{F}$.

For buck mode,



For boost mode,



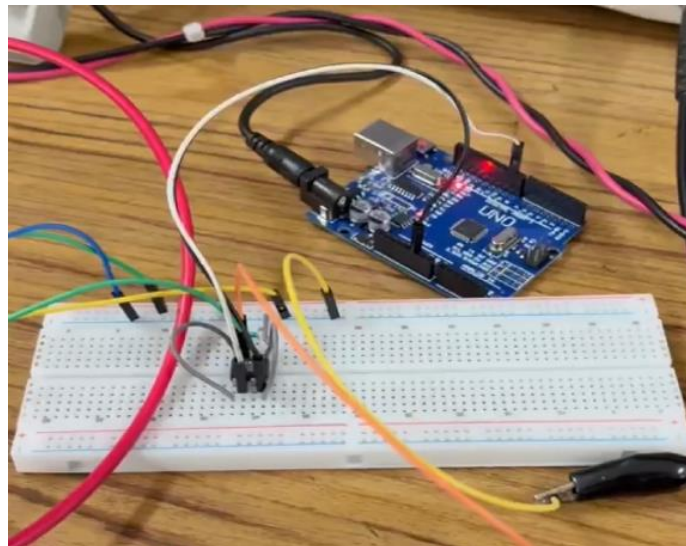
As we can see, we have again gotten the correct operation. Although the output voltage isn't exactly equal to the target, in actual physical experiment we have varied the duty cycle a bit and got proper results.

3.5 Hardware Implementation

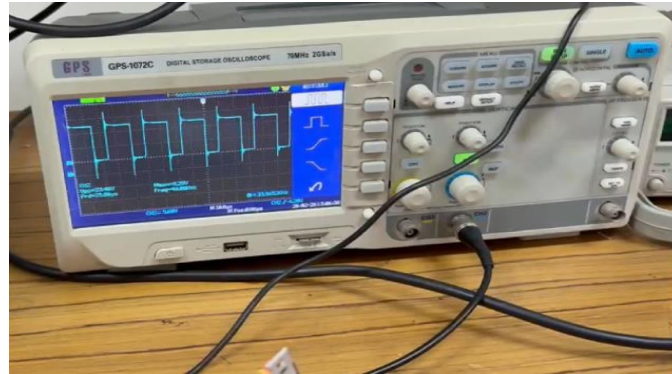
PWM Generation

First, we have generated the duty cycle using Arduino and used it at the gate of the MOSFET through the gate driver circuit MIC4422bn.

Circuit:



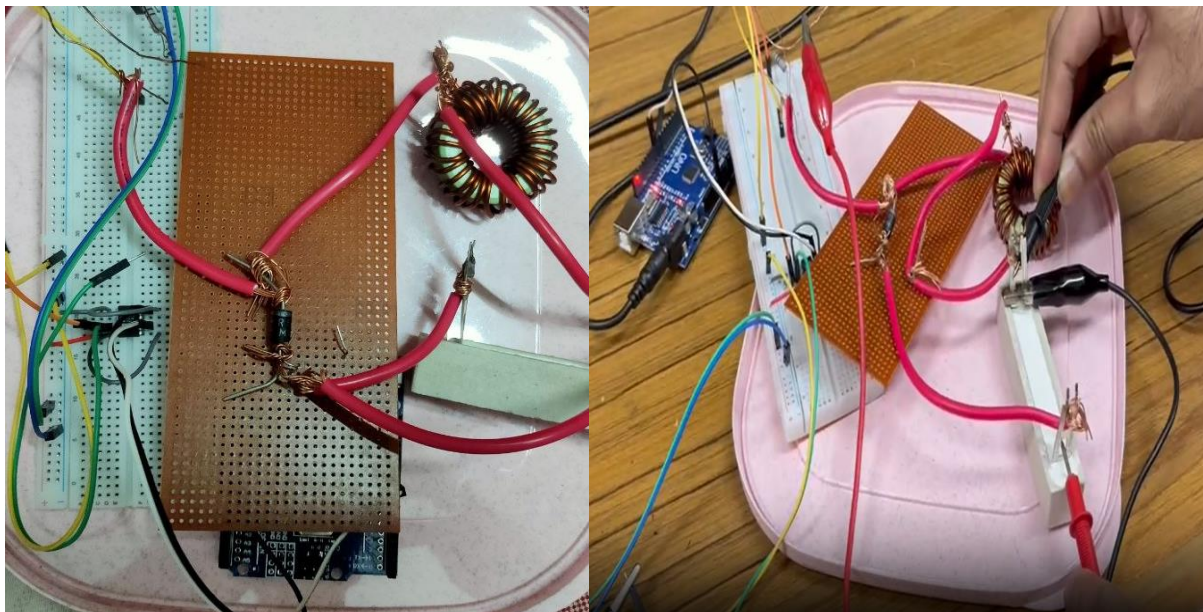
Output:



We get 0V to 10V pulse to drive the MOSFET gate with our specific duty cycle.

For the main circuit, we used end to end connection between components and utilized 10 AWG wire instead of typical jumper wires.

We could work with breadboard only in case of buck mode operation. For boost mode, we needed to use Veroboard. Moreover, Veroboard is well suited for both buck and boost mode.



4 Implementation

4.1 Description

The implementation of the buck–boost converter involves the practical realization of the designed circuit using discrete components, followed by testing and validation of its performance in both buck and boost modes. The system is constructed by integrating the power stage, control unit, and driver circuitry to achieve controlled voltage conversion from a 12V DC input.

The overall implementation can be divided into three main parts: PWM generation, gate driving, and power conversion stage.

PWM Generation

The PWM control signal is generated using an Arduino microcontroller. The Arduino is programmed to produce a square wave signal with a fixed switching frequency of approximately 40 kHz.

The duty cycle of this signal is adjusted depending on the mode of operation:

- For buck mode, a duty cycle of approximately 30–35% is applied to obtain an output voltage close to 5V.
- For boost mode, a duty cycle of approximately 60–65% is used to achieve an output voltage near 18V.

The PWM signal acts as the control input of the system and determines the amount of energy transferred from the input to the output.

Gate Driver Circuit

The PWM output from the Arduino is fed into the MIC4422 MOSFET driver IC. This driver amplifies the control signal and provides sufficient current to rapidly charge and discharge the gate capacitance of the MOSFET (IRFZ44N).

This stage ensures:

- Fast switching transitions
- Reduced switching losses
- Improved efficiency and reliability

Without the driver IC, the MOSFET would switch slowly, leading to excessive heat generation and reduced performance.

Power Conversion Stage

The power stage consists of the MOSFET, inductor, diode, capacitor, and load resistor arranged in a buck–boost configuration.

- When the MOSFET is turned ON, current flows through the inductor, storing energy in its magnetic field.
- When the MOSFET is turned OFF, the inductor releases its stored energy through the Schottky diode to the output capacitor and load.

The output capacitor smooths the pulsating voltage and provides a nearly constant DC output to the load.

The inductor used in the circuit is wound on a T141-52 iron powder toroidal core, ensuring efficient energy storage and minimal core losses. The Schottky diode (SR560) enables fast switching and reduces conduction losses, improving overall efficiency.

Hardware Assembly

The circuit is assembled on a breadboard/prototyping board using proper wiring and component placement. Care is taken to:

- Keep high-current paths short to reduce losses
- Ensure proper grounding
- Minimize noise and interference

The load is implemented using a 10 Ω , 50W aluminium housing resistor to safely dissipate power during operation.

A regulated laboratory DC power supply is used to provide a stable 12V input to the system.

5 Design Analysis and Evaluation

5.1. Novelty

Although the buck-boost converter topology itself is a well-known DC–DC conversion method, the distinct contribution of this project lies in its holistic implementation, which combines traditional analog circuit design with contemporary microcontroller-driven digital control to form a complete end-to-end system.

A key innovative aspect of this work is the use of a microcontroller-based PWM control (Arduino) combined with a dedicated high-speed MOSFET driver (MIC4422). This approach allows flexible and precise control of the duty cycle, enabling seamless switching between buck and boost modes without requiring hardware reconfiguration. Such flexibility is particularly beneficial in applications where input conditions vary or multiple output voltage levels are required.

Another important contribution is the custom design and fabrication of the inductor using an iron powder toroidal core (T141-52). Instead of relying on commercially available inductors, the inductor is designed based on theoretical calculations and manually wound, demonstrating an in-depth understanding of magnetic component design and its impact on converter performance.

5.2. Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

The design and implementation of the buck–boost converter have been carried out with careful consideration of public health and safety aspects, particularly due to the involvement of electrical power, high current levels, and heat-generating components.

One of the primary safety concerns is electrical hazard. Although the system operates at a relatively low voltage (12V), the current levels—especially in boost mode—can be significant. Proper insulation of wires, secure connections, and avoidance of exposed conductive parts are ensured to prevent accidental short circuits or electric shocks.

Another critical factor is thermal safety. Components such as the MOSFET, diode, and load resistor dissipate considerable power during operation. The use of a high-power (50W) aluminium housing resistor helps in effective heat dissipation. Additionally, components are operated within their rated limits to prevent overheating, which could otherwise lead to component failure or fire hazards.

During laboratory testing, standard safety practices are followed, such as using regulated power supplies with current limiting features.

Finally, the design ensures that all components are used within their rated voltage, current, and power limits, reducing the risk of catastrophic failure.

5.2.2 Considerations to Environment

The buck-boost converter promotes environmental sustainability by improving energy efficiency, thereby reducing power losses and unnecessary energy consumption. Efficient DC-DC conversion minimizes heat generation, which lowers cooling requirements and extends component lifespan, reducing electronic waste. Furthermore, the use of compact and optimized components contributes to reduced material usage, aligning with environmentally responsible engineering practices.

5.2.3 Considerations to Cultural and Societal Needs

The developed buck-boost converter supports societal needs by enabling reliable and affordable power conversion for everyday electronic devices, especially in developing regions. Its ability to both

step up and step down voltage efficiently makes it suitable for many applications like portable consumer electronics, renewable energy systems, electric vehicles and hybrid vehicles. This contributes to technological advancement and supports digital inclusion in diverse cultural and socioeconomic environments.

5.3. Investigations

5.3.1 Data Collection

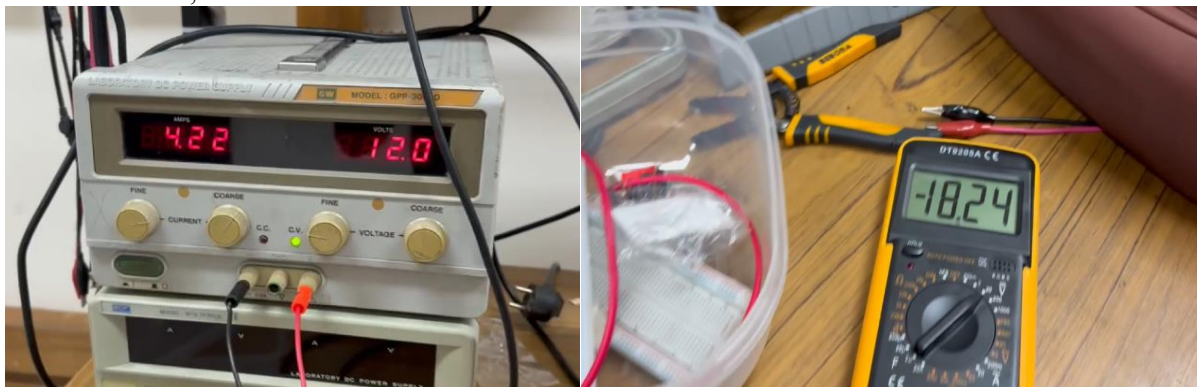
For buck mode,



In the left picture, we can see the input current is **0.36 A** and input voltage is **12 V**.

In the right picture, we can see the output voltage is **4.98 V** which is almost equal to our desired 5V output voltage.

For boost mode,



In the left picture, we can see the input current is **4.22 A** and input voltage is **12 V**.

In the right picture, we can see the output voltage is **18.24 V** which is almost equal to our desired 18V output voltage.

5.3.2 Results and Analysis

Duty cycle

We needed to use duty cycle 0.35 instead of the theoretically ideal duty cycle of 0.3 for buck mode and for boost mode we needed to 0.65 in place of 0.6.

The ideal equation for duty cycle is

$$k = \frac{V_o}{V_o + V_s}$$

But taking into consideration inductor and capacitor resistance, we get

$$V_o = \frac{k}{1-k} \left[\frac{(1-k)^2 R}{(1-k)^2 R + r_L + k(1-k) \left(\frac{r_c R}{r_c + R} \right)} \right] V_s$$

Here, we have inductor's resistance value. But we don't know the resistance of the capacitor. So, we take it as 0.

The equation turns out to be

$$V_o = \frac{k}{1-k} \frac{(1-k)^2 R}{(1-k)^2 R + r_L} V_s$$

We have $R=10 \text{ ohm}$, $r_L=0.3 \text{ ohm}$, $V_s=12\text{V}$.

For buck mode,

$$V_o=5\text{V},$$

Solving the equation, we get

$$k = 0.31$$

So, it remains pretty much the same.

We used $k=0.35$.

So, the error in k is

$$\frac{0.35 - 0.31}{0.31} = 12.9\%$$

For boost mode,

$$V_o=18\text{V},$$

Solving the equation, we get

$$k = 0.652$$

Here, we see a significant change.

The ideal k was 0.6 and our used k was 0.65.

So, the error in k is

$$\frac{0.652 - 0.65}{0.652} = 0.3\%$$

Buck mode—

Efficiency and power loss analysis

Output Voltage

$$V_o = 4.97$$

Output current

$$I_o = \frac{4.97}{10} = 0.497$$

Output power

$$P_o = \frac{V_o^2}{R} = 2.47$$

Input current

$$I_s = 0.36$$

Input power

$$P_{in} = V_s I_s = 4.32$$

Efficiency

$$\eta = \frac{P_o}{P_{in}} \times 100$$
$$= 57.2 \%$$

Average inductor current

$$I_{L,avg} = \frac{I_o}{1 - k} = 0.765$$

We assume the form factor is 1.15.

So, the rms inductor current

$$I_{L,rms} = I_{L,avg} \times FF = 0.88 \text{ A}$$

Inductor power loss

$$P_{ind} = I_{L,rms}^2 r_L = 0.23 \text{ W}$$

Average diode current

$$I_{D,avg} = I_o = 0.497 \text{ A}$$

Diode forward bias voltage

$$V_{D,FB} = 0.7 \text{ V}$$

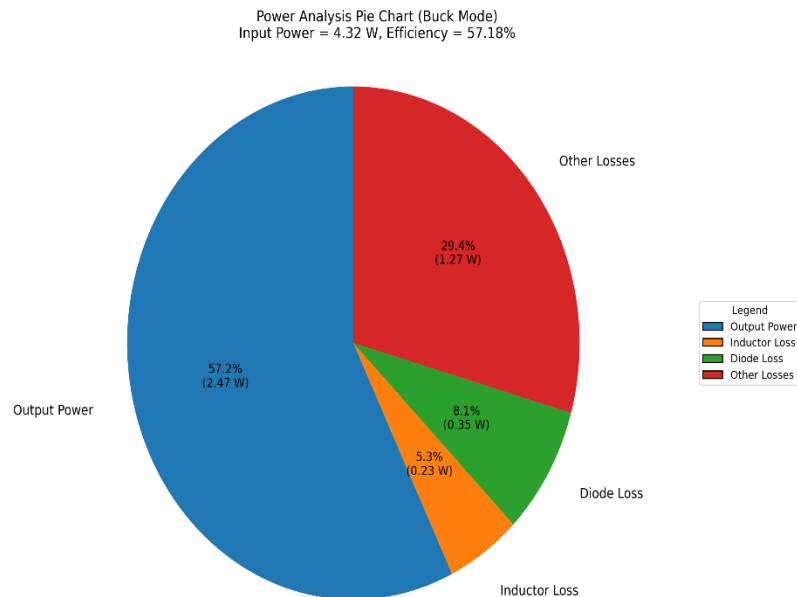
Diode power loss

$$P_{diode} = I_{D,avg} V_{D,FB} = 0.35 \text{ W}$$

So, for buck mode diode power loss is greater than inductor power loss. But still there are significant other losses. And a major factor maybe switching loss.

Other Loss

$P_{in} - P_{out} - \text{inductor loss} - \text{diode loss} = 1.27 \text{ W}$



Boost Mode—

Efficiency and power loss analysis

Output Voltage

$$V_o = 18.24 \text{ V}$$

Output current

$$I_o = \frac{18.24}{10} = 1.824$$

Output power

$$P_o = \frac{V_o^2}{R} = 33.27 \text{ W}$$

Input current

$$I_s = 4.22 \text{ A}$$

Input power

$$\begin{aligned} P_{in} &= V_s I_s \\ &= 50.64 \text{ W} \end{aligned}$$

Efficiency

$$\begin{aligned} \eta &= \frac{P_o}{P_{in}} \times 100 \\ &= 65.7 \% \end{aligned}$$

Average inductor current

$$I_{L,avg} = \frac{I_o}{1-k} = 5.211 \text{ A}$$

We assume the form factor is 1.15.

So, the rms inductor current

$$I_{L,rms} = I_{L,avg} \times FF = 6 \text{ A}$$

Inductor power loss

$$P_{ind} = I_{L,rms}^2 r_L = 10.8 \text{ W}$$

Average diode current

$$I_{D,avg} = I_o = 1.824 \text{ A}$$

Diode forward bias voltage

$$V_{D,FB} = 0.7 \text{ V}$$

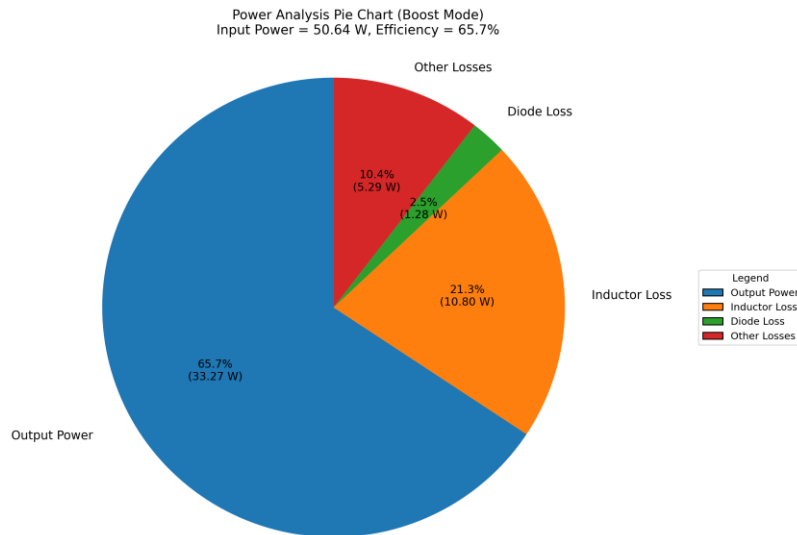
Diode power loss

$$P_{diode} = I_{D,avg} V_{D,FB} = 1.28 \text{ W}$$

So, for boost mode inductor power loss is greater than diode power loss. But still there are significant other losses. And a major factor maybe switching loss.

Other Loss

$P_{in} - P_{out} - \text{inductor loss} - \text{diode loss} = 5.29 \text{ W}$



5.3.3 Interpretation and Conclusions on Data

From our analysis we can see that although in principle we get accurate behaviour in both buck and boost modes. The exact result often doesn't accurately match the theoretical ideal model.

The experimental results and analytical calculations provide important insights into the practical performance of the buck-boost converter and highlight the deviations from ideal theoretical behavior. One of the key observations from the data is the variation in required duty cycle compared to theoretical predictions.

For the buck mode, the calculated duty cycle considering losses is approximately 0.31, which is close to the ideal value of 0.3. Experimentally, a duty cycle of 0.35 was required to achieve the desired output voltage, resulting in an error of approximately 12.9%.

In contrast, for boost mode, the effect of non-idealities is more significant. The corrected duty cycle becomes 0.652 instead of the ideal 0.6, which closely matches the experimentally used value of 0.65, with a very small error of 0.3%. This indicates that non-ideal effects have a greater impact in boost operation, primarily due to higher current levels and increased conduction losses.

The efficiency analysis further supports these observations. In buck mode, the efficiency is approximately 57.2%, whereas in boost mode it increases to around 65.7%. Although boost mode handles higher power, its efficiency remains higher due to better energy transfer characteristics at higher duty cycles, despite increased losses.

- Buck Mode:

The diode power loss (0.35 W) is greater than the inductor loss (0.23 W), indicating that the forward voltage drop of the diode plays a significant role in reducing efficiency. However, the total measured losses exceed the sum of these values, suggesting additional contributions from switching losses, MOSFET conduction losses, and parasitic effects.

- Boost Mode:

The inductor loss (10.8 W) is significantly higher than the diode loss (1.28 W). This is due to the large inductor current (approximately 6 A RMS), which increases copper losses. This clearly shows that inductor losses dominate in high-power boost operation, making inductor design a critical factor in efficiency optimization.

Another important observation is the role of switching losses, which are not explicitly calculated but inferred from the difference between total input power and the sum of calculated losses. At a switching frequency of 40 kHz, the MOSFET experiences frequent transitions, leading to energy loss during turn-on and turn-off periods.

From the analysis, it can be concluded that the buck–boost converter successfully achieves the desired voltage conversion in both modes, with experimental results closely aligning with theoretical expectations when non-idealities are considered.

The study highlights the importance of including practical factors such as component resistances, switching losses, and real operating conditions in the design process. It also emphasizes that improving efficiency requires careful optimization of key components, particularly the inductor and switching elements.

Despite moderate efficiency, the converter demonstrates reliable operation and validates the effectiveness of the design methodology. The insights gained from this analysis provide a strong foundation for further improvements, such as enhanced component selection, better thermal management, and advanced control strategies.

5.4. Limitations of Tools (PO(e))

The design, simulation, and implementation of the buck–boost converter rely on several hardware and software tools, each of which has inherent limitations that affect the accuracy, performance, and reliability of the final system.

One of the primary tools used in this project is Proteus for simulation. While simulation provides an idealized environment for analyzing converter behavior, it often neglects non-ideal factors such as parasitic resistances, switching delays, electromagnetic interference, and thermal effects and inductor resistance. As a result, simulation results tend to be more optimistic compared to real-world performance. This discrepancy was observed in the project, where the practical efficiency was lower than simulated values.

The Arduino microcontroller used for PWM generation also has certain limitations. It cannot generate extremely precise PWM signals due to clock resolution constraints. Additionally, the output current capability of the Arduino pins is limited, which necessitates the use of an external driver IC. The lack of built-in high-resolution PWM control may introduce some inaccuracies in duty cycle generation, affecting output voltage regulation.

The MOSFET driver IC (MIC4422), although effective, has finite switching speed and propagation delay. These delays can lead to switching losses, especially at higher frequencies. Moreover, the driver requires proper power supply decoupling; otherwise, it may introduce noise or unstable switching behavior.

Another significant limitation arises from component tolerances. Real components such as inductors, capacitors, and resistors do not have exact values as specified. Variations in inductance, equivalent series resistance (ESR), and capacitance can influence ripple, efficiency, and stability of the converter.

Finally, thermal effects are not fully captured by design tools. As components heat up during operation, their electrical characteristics change, which may lead to deviations from expected performance.

5.5. Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The buck–boost converter has significant relevance in modern society due to its wide range of applications in consumer electronics, renewable energy systems, and portable devices. This project demonstrates a technology that can contribute to improving energy accessibility and efficiency, particularly in developing regions.

In many parts of the world, including regions with unstable power supply, voltage fluctuations are common. A buck–boost converter can help maintain stable voltage levels for sensitive electronic devices, thereby improving reliability and lifespan. This is especially important for educational institutions, healthcare facilities, and small-scale industries where consistent power is essential.

From a cultural perspective, the adoption of such technologies promotes technological awareness and self-reliance. Designing and implementing a converter using locally available components encourages hands-on learning and innovation among students and engineers. It supports a culture of practical engineering and problem-solving rather than dependence on imported solutions.

5.5.2 Assessment of Health and Safety Issues

One of the primary concerns is electrical safety. Although the operating voltage is relatively low, high current levels can still pose risks such as short circuits, sparks, and component damage. Improper handling or faulty connections may lead to accidents or equipment failure.

Another important issue is thermal safety. Components such as the MOSFET, diode, and load resistor generate heat during operation. Excessive heating can cause burns, degrade component performance, or even result in hazards if not properly managed. The use of high-power rated components and proper heat dissipation mechanisms helps mitigate these risks.

5.5.3 Assessment of Legal Issues

The design and implementation of a buck–boost converter also involve certain legal considerations, particularly when transitioning from a prototype to a commercial product.

One of the key legal aspects is compliance with electrical standards and regulations. Any power electronic device intended for commercial use must meet safety and performance standards set by regulatory authorities. These may include standards related to electrical insulation, electromagnetic compatibility (EMC), and energy efficiency.

5.6 Sustainability Evaluation (PO(g))

One of the key contributions to sustainability is energy efficiency. By converting DC voltage levels efficiently, the converter minimizes energy wastage compared to linear regulators, which dissipate excess energy as heat. Although the prototype shows moderate efficiency, the design concept itself supports improved energy utilization, especially when optimized further.

The use of reusable and durable components, such as MOSFETs, toroidal inductors, and aluminium-housed resistors, contributes to longer operational life and reduces the need for frequent replacement. This helps in minimizing electronic waste over time.

5.7 Ethical Issues (PO(h))

One of the primary ethical responsibilities is ensuring accuracy and honesty in reporting results. All experimental data, including efficiency values and output measurements, are presented transparently,

even when they deviate from theoretical expectations. This reflects integrity in engineering practice and avoids misleading conclusions.

The project also considers responsible use of resources. Components are selected based on availability and cost-effectiveness, avoiding unnecessary waste.

Furthermore, the project avoids any form of plagiarism or misrepresentation, ensuring that all work reflects the actual contributions of the team members.

6 Reflection on Individual and Team Work

6.1 Individual Contribution of Each member

Student ID	Name	Accomplished Tasks
2106003	Rejoun Rahi Rad	Component Ordering, Purchasing and Collection, Boost mode circuit implementation, Report Writing
2106005	Abrar Fahim	Simulation on proteus, Arduino Coding for PWM generation, Buck Mode circuit implementation, PPT preparation
2106011	Md. Yeasir Alif	Simulation on Simulink, Mic4422Bn MOSFET driver circuit implementation and checking, Report Writing
2106012	Al Muktadir Khan	Circuit designing and theoretical calculation, Inductor designing, wire winding on core, Inductor value checking, Report Writing

6.2 Log Book of Project Implementation

Date	Activity / Progress
December 23	The initial project idea was discussed and finalized within the group.
January 17	Theoretical calculation and software simulation was done
January 26	PWM Generation Circuit was successfully implemented with Arduino and MOSFET driver IC
February 9	Buck mode was successfully implemented in hardware using breadboard
February 10	Boost mode was implemented in breadboard but it didn't work
February 16	Boost mode was successfully implemented using veroboard
February 17	Both Buck and Boost mode implemented with finetuning, proper video recording and mathematical analysis of the final result were completed

7 Communication to External Stakeholder (PO(j))

7.1 Executive Summary

This project presents the design and implementation of a buck–boost converter capable of converting a 12V DC input into both lower (5V) and higher (18V) output voltages. The system uses a programmable microcontroller to control switching and regulate voltage efficiently. A hardware prototype was developed and tested, showing reliable performance in both modes. This technology is widely applicable in power electronics, especially in battery-powered devices and renewable energy systems where voltage levels vary. The project demonstrates a cost-effective and flexible solution for voltage regulation, highlighting its importance in modern electronic applications.

7.2 User Manual

This section provides a step-by-step guide for operating the buck–boost converter safely and effectively.

a) Required Equipment

Before operating the system, ensure the following components are available:

- DC power supply (set to 12V)
- Buck–boost converter circuit (assembled)
- Arduino microcontroller (programmed)
- MOSFET driver circuit (MIC4422)
- Load resistor (10 Ω , 50W)
- Multimeter (for voltage/current measurement)

b) Initial Setup

- 1 Connect the DC power supply to the input terminals of the converter circuit.
- 2 Ensure correct polarity (positive to positive, ground to ground).
- 3 Connect the load resistor across the output terminals.
- 4 Verify all circuit connections, especially:
 - MOSFET orientation
 - Diode polarity
 - Inductor connections
- 5 Connect the Arduino to a power source (external supply).

c) Operating Procedure

Step 1: Power ON

- Turn on the DC power supply and set it to 12V.
- Ensure the Arduino is powered and running the PWM program.

Step 2: Select Mode of Operation

- The mode is determined by the duty cycle programmed in the Arduino:
- Buck Mode (5V Output):

Set duty cycle $\approx$ 30–35%

- Boost Mode (18V Output):

Set duty cycle $\approx$ 60–65%

Step 3: Observe Output

- Measure output voltage using a multimeter.
- Verify that:
 - Output $\approx$ 5V in buck mode
 - Output $\approx$ 18V in boost mode

d) Monitoring and Adjustment

- If the output voltage deviates:
- Adjust the PWM duty cycle slightly
- Check for loose connections

- Monitor temperature of:
 - MOSFET
 - Diode
 - Load resistor

e) Safety Precautions

- Always turn OFF power before modifying connections
- Operate within rated voltage and current limits

7.3 YouTube Link

[AFN2760/BuckBoostConverter](#)

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

The following table lists the price of the components required to build the prototype.

Serial	Item	Quantity	Unit Price (BDT)	Total Price (BDT)
1	SR560 5A 60V Schottky Rectifier Diode	2	12	24
2	10 ohm 50W Resistor	2	143	286
3	T141-52 Ferrite Ring Toroidal Inductor Iron Powder Core	1	140	140
4	Breadboard Jumper Wire Set	1	190	190
5	16 SWG Enameled Pure Copper Wire Magnet Wire (100 gram)	1	244	244
6	Veroboard Dot Type	1	35	35
7	Arduino Uno R3 SMD	1	470	470
8	IRFZ44N MOSFET	2	55	110
9	Capacitor 470uF 50V	2	8	16
10	Heat Sink	1	25	25
11	10 AWG wire	2	70	140
Total				1680

The table on the previous page takes a grand total of Tk 1680. Apart from the bill of materials listed above, we need to add another Tk 200 as transportation and shipping cost. Therefore, per unit cost of prototype is Tk **1880**.

9 Future Work (PO(i))

Although the current implementation of the buck–boost converter demonstrates successful operation, several improvements and extensions can be made to enhance its performance, efficiency, and practical applicability.

One major area for future work is the implementation of a closed-loop feedback control system. In the current design, the duty cycle is manually set, which may lead to variations in output voltage under changing load or input conditions. By incorporating voltage feedback and a control algorithm (such as PID control), the system can automatically regulate the output voltage and maintain stability.

The design can also be extended by developing a PCB-based implementation instead of a veroboard setup. This would reduce parasitic effects, improve reliability, and make the system suitable for real-world applications.

Another potential extension is the integration of the converter into renewable energy systems, such as solar panels or battery management systems, where variable voltage conversion is essential.

Additionally, advanced features such as protection mechanisms (overvoltage, overcurrent, thermal

protection) can be incorporated to make the system more user-friendly and robust.

Finally, further research can focus on improving efficiency and reducing ripple, as well as exploring advanced converter topologies such as SEPIC or Cuk converters for enhanced performance.

10 References

- [1] Power Electronics Devices, Circuits and Application (4th Edition) by Muhammad H. Rashid
- [2] Power Electronics by Daniel W. Hart
- [3] EEE 316 Lab Experiment: 03
- [4] AN-2579: The Design of the Inverting Buck/Boost Converter Topology
<https://www.analog.com/en/resources/app-notes/an-2579.html>
- [5] A Buck-Boost Converter; Design, Analysis and Implementation Suggested for Renewable Energy Systems
https://www.researchgate.net/publication/342232210_A_Buck-Boost_Converter_Design_Analysis_and_Implementation_Suggested_for_Renewable_Energy_Systems
- [6] An enhanced buck-boost converter for photovoltaic diagnosis application: Accurate MPP tracker and I-V tracer
<https://www.sciencedirect.com/science/article/pii/S2468227625000328>